Chapter 4
Basic Landscape Analysis

Tangible Landscape allows us to explore the spatial patterns of topographic parameters and their relation to basic surface geometry. We can easily sculpt a physical model of a landscape with our hands changing its topography. With Tangible Landscape we can analyze the topography of a landscape model and how it changes by continually 3D scanning the model and computing DEMs from the scanned point clouds using binning or interpolation. By computing basic topographic parameters, morphometric units, and DEM differencing we can map changes in elevation, slope, and landform as the model is modified. These maps are then projected over the physical model of the landscape so that we have near real-time feedback and can understand the impact of the changes we are making as we make them.

4.1 Processing and Analyzing the Scanned DEM

To start working with Tangible Landscape, the setup is first calibrated using a flat surface, typically a table, to minimize the scanning distortion (see Sect. 2.2.4). After successful calibration the model can be scanned continuously. With each scan the edges of the model are detected and georeferenced and the point cloud is converted into a DEM via binning or interpolation. Standard GRASS GIS modules are then used to compute maps of topographic parameters or landforms that are then projected over the model. This chapter explains how the DEM is derived and demonstrates the results in a case study.

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4.1.1 Creating DEM from Point Cloud

There are two ways to derive a raster DEM from a set of points in Tangible Landscape:

Binning A cell value is assigned based on the univariate statistics of the points that fall inside the particular cell. When constructing a DEM the mean of the z coordinates is typically used. Binning is fast, but creates a rough surface so it is only useful for certain types of analyses.

Interpolation Cell values are estimated from available point data using a selected mathematical or statistical interpolation technique.

4.1.2 Interpolation with the RST Function

The point clouds produced by the scanning process are usually noisy. The regularized spline with tension (RST) function, implemented as the *v.surf.rst* module in GRASS GIS, can be used to interpolate smooth DEMs from noisy point data and simultaneously compute topographic parameters such as slope and curvature. RST approximates a surface from data points by minimizing a smoothness seminorm and the deviations between the given points and the resulting surface (Mitasova et al. 2005). The RST smoothness seminorm includes derivatives of all orders with their weights decreasing with the increasing derivative order leading to the function:

$$z(\mathbf{r}) = a_1 + \sum_{j=1}^{N} \lambda_j R(\varrho_j) \tag{4.1}$$

$$R(\varrho_j) = -[E_1(\varrho_j) + \ln(\varrho_j) + C_E] \tag{4.2}$$

where $z(\mathbf{r})$ is the elevation at a point $\mathbf{r} = (x, y)$, a_1 is a trend, λ_j are coefficients, N is a number of given points, $R(\varrho_j)$ is a radial basis function, $\varrho_j = (\varphi r_j/2)^2$, φ is a generalized tension parameter, $r_j = |\mathbf{r} - \mathbf{r}_j|$ is a distance, $C_E = 0.577215$ is the Euler constant, and $E_1(\varrho_j)$ is the exponential integral function (Abramowitz and Stegun 1965; Mitasova and Mitas 1993). The coefficients a_1, $\{\lambda_j\}$ are obtained by solving the system of linear equations:

$$\sum_{j=1}^{N} \lambda_j = 0 \tag{4.3}$$

$$a_1 + \sum_{j=1}^{N} \lambda_j \left[R(\varrho_j) + \delta \frac{w_0}{w_j} \right] = z(\mathbf{r}_i), \qquad i = 1, \ldots, N \tag{4.4}$$

where w_0/w_j are positive weighting factors representing a smoothing parameter at each given point $\mathbf{r_j} = (x_j, y_j)$.

Tension influences the detail of the surface and smoothing influences the deviations between the given points and the resulting surface. A higher smoothing value can be used to reduce the noise in the data. Because this is a global operation acting on the entire dataset it can be processing intensive for large datasets so the *v.surf.rst* implementation uses a quadtree segmentation algorithm with smooth overlaps to separately process overlapping neighborhoods of points. A quadtree is a tree data structure that recursively partitions processing regions into smaller processing regions by dividing them into quadrants. Depending on the distribution of the input points in relation to the surface complexity, interpolation with *v.surf.rst* may require experimental, iterative tuning of the parameters. By adjusting the parameters which control the RST function properties and distribution of points used for interpolation, this function can be used to accurately model smooth or rough topography (Mitasova and Mitas 1993; Mitasova et al. 2005).

4.1.3 Analyzing the DEM

The geometry of an elevation surface at any point can be described by topographic parameters—slope, aspect, and several types of curvatures (Mitasova et al. 2005; Olaya 2009). Elevation surface can be partitioned into units with specific geometric properties; one approach to defining these units is geomorphons.

Basic Topographic Parameters The steepest slope angle β in degrees or percent and the aspect angle α in degrees are the most commonly used topographic parameters topographic parameters (Fig. 4.2). They represent the magnitude and direction of the surface gradient vector $\nabla z = (f_x, f_y)$ (its direction is upslope) and are computed as follows:

$$\beta = \arctan\sqrt{f_x^2 + f_y^2} \qquad \beta[\%] = 100\sqrt{f_x^2 + f_y^2} \tag{4.5}$$

$$\alpha = \arctan -\frac{f_y}{f_x} \tag{4.6}$$

where $f_x = \partial z/\partial x$ and $f_y = \partial z/\partial y$ are the first order partial derivatives of elevation surface function $z = f(x, y)$. The slope values range from 0° to 90° and the aspect values range from 0° to 360°. Therefore, computing the correct angle for aspect requires the evaluation of all possible combinations of negative, positive and zero values of f_x, f_y in relation to the selected direction of 0° (usually east or north).

For applications in geosciences the curvature in the gradient direction (profile curvature) is important because it reflects the change in the slope angle and thus controls the change of the velocity of mass flowing downwards along the slope curve. The equation for the profile curvature is

$$\kappa_s = \frac{f_{xx}f_x^2 + 2f_{xy}f_xf_y + f_{yy}f_y^2}{p\sqrt{q^3}} \tag{4.7}$$

where κ_s is the profile curvature in m^{-1} and f_{xx}, f_{xy}, f_{yy} are the second order partial derivatives of the elevation surface function $z = f(x, y)$, $p = f_x^2 + f_y^2$ and $q = p + 1$. The curvature in a direction perpendicular to the gradient (tangential curvature) reflects the change in the aspect angle and influences the divergence/convergence of water flow. The equation for tangential curvature at a given point is

$$\kappa_t = \frac{f_{xx}f_y^2 - 2f_{xy}f_xf_y + f_{yy}f_x^2}{p\sqrt{q}} \tag{4.8}$$

where κ_t is the tangential curvature in m^{-1}. Partial derivatives can be computed from a suitable approximation function $z = f(x, y)$ such as local second order polynomial or certain types of splines.

Partial Derivatives from a Raster DEM We can use a second order polynomial approximation of a surface defined by a given grid point and its 3×3 neighborhood to estimate the topographic parameters (Horn 1981):

$$z(x, y) = a_0 + a_1x + a_2y + a_3xy + a_4x^2 + a_5y^2 \tag{4.9}$$

By fitting this polynomial to the nine grid points (the given point $z_{i,j}$ and its 3×3 neighborhood as shown below in Fig. 4.1) using weighted least squares we can derive the coefficients of this polynomial as well as its partial derivatives ($f_x = a_1$, $f_y = a_2$, $f_{xx} = 2a_4$, $f_{yy} = 2a_5$, $f_{xy} = a_3$):

$$f_x = \frac{(z_{i-1,j-1} - z_{i+1,j-1}) + 2(z_{i-1,j} - z_{i+1,j}) + (z_{i-1,j+1} - z_{i+1,j+1})}{8\Delta x} \tag{4.10}$$

$z_{i-1,j+1}$	$z_{i,j+1}$	$z_{i+1,j+1}$
$z_{i-1,j}$	$z_{i,j}$	$z_{i+1,j}$
$z_{i-1,j-1}$	$z_{i,j-1}$	$z_{i+1,j-1}$

Fig. 4.1 The elevation values of a grid cell and its 3×3 neighborhood

$$f_y = \frac{(z_{i-1,j-1} - z_{i-1,j+1}) + 2(z_{i,j-1} - z_{i,j+1}) + (z_{i+1,j-1} - z_{i+1,j+1})}{8\Delta y} \tag{4.11}$$

where Δx and Δy is the resolution (grid spacing) in the east-west and north-south directions respectively. To compute the second order partial derivatives we first denote $D(i,\delta) = z_{i,j+1} + z_{i,j-1} - 2z_{i,j}$ and $D(\delta,j) = z_{i+1,j} + z_{i-1,j} - 2z_{i,j}$. Then we can write:

$$f_{xx} = \frac{D(\delta,j+1) + (4z_{i-1,j} + 4z_{i+1,j} - 8z_{i,j}) + D(\delta,j-1)}{6(\Delta x)^2} \tag{4.12}$$

$$f_{yy} = \frac{D(i-1,\delta) + 4(z_{i,j+1} + z_{i,j-1} - 2z_{i,j}) + D(i+1,\delta)}{6(\Delta y)^2} \tag{4.13}$$

$$f_{xy} = \frac{(z_{i-1,j-1} - z_{i+1,j-1}) - (z_{i-1,j+1} - z_{i+1,j+1})}{4\Delta x\Delta y} \tag{4.14}$$

where $z_{i,j}$ is the elevation value at row j and column i, Δx is the east-west grid spacing, and Δy is the north-south grid spacing (resolution).

Landforms Landforms can be identified automatically by fitting a quadratic function to the elevation values in a given neighborhood or 'moving window' using least squares. This method for identifying landforms using differential geometry is implemented in GRASS GIS as a module *r.param.scale* (Fig. 4.3). While real-world landforms are scale-dependent and may be nested, this method can only identify landforms at a single scale based on the size of the moving window.

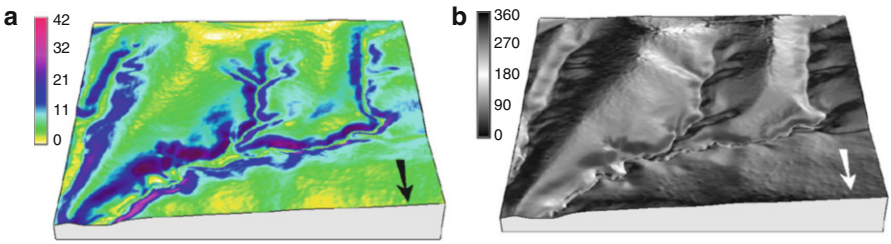

Fig. 4.2 Slope (**a**) and aspect (**b**) maps in degrees draped over a DEM

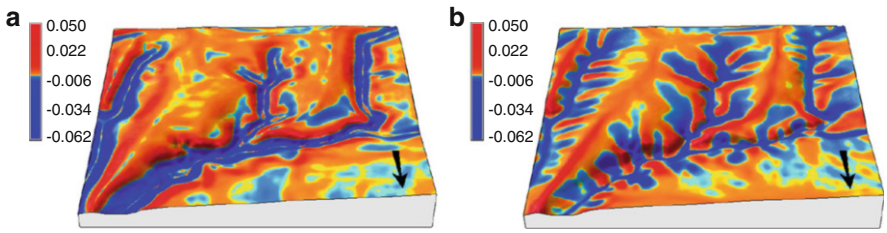

Fig. 4.3 Profile (**a**) and tangential (**b**) curvature maps draped over a DEM

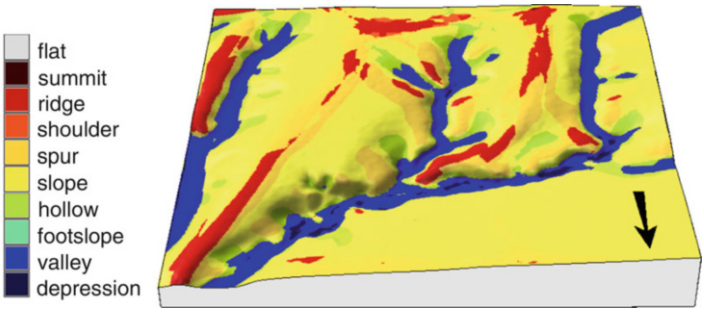

Fig. 4.4 Landforms identified by geomorphons (see *r.geomorphon* manual page for explanation of landform types)

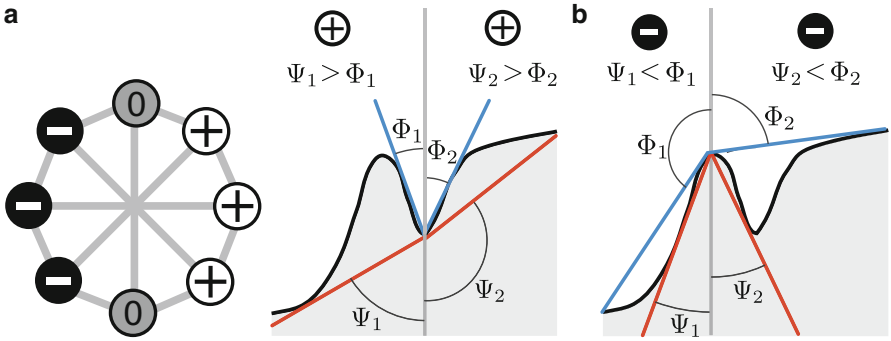

Fig. 4.5 Geomorphons: local ternary patterns and line-of-sights concepts applied to landform identification. The relationship between a cell and its neighbors in terms of relative elevation (higher than, equal to, or lower than) is expressed as a ternary pattern (**a**). Ternary patterns are derived using the line-of-sight method and comparing the zenith angle Φ and nadir angle Ψ (**b**)

Geomorphons—geomorphologic phonotypes—is a novel method for identifying landforms using pattern recognition developed by Jasiewicz and Stepinski (2013) and implemented in GRASS GIS as the add-on module *r.geomorphon* (Fig. 4.4). A geomorphon is an abstract unit of terrain described by the local ternary pattern rather than by relief. The relationship between a cell and its neighborhood is described using an 8-tuple pattern of lower (−), equal (0), higher (+) elevation values (Fig. 4.5a). Neighborhoods are based on visibility rather than the predefined dimensions of a moving window. Geomorphons uses the line-of-sight principle to dynamically determine the optimal size of the search radius for a neighborhood based on openness of the terrain (see Fig. 4.5b). Thus, unlike methods based on differential geometry, the geomorphons approach is able to efficiently classify landforms across a range of spatial scales (Jasiewicz and Stepinski 2013).

4.2 Case Study: Topographic Analysis of Graded Landscape

In this section we demonstrate how changes in elevation impact topographic parameters and landforms in a small study area on NCSU's Centennial Campus.

4.2.1 Site Description and Model

Our 56 ha case study area is located on NCSU's Centennial Campus on the southwest side of Lake Raleigh (Fig. 4.6). The topography is largely natural, but there are some anthropogenic features, such as informal trails and the Chancellor's house with its access road, parking, and retention pond. There is rapid development on the campus; several new buildings are currently under construction. At the same time there is a desire to preserve part of the campus as a park and a natural teaching area. In this case study we developed part of the natural area as student housing and constructed an access road. Since significant grading would be required to develop the site given the steep slopes we extensively modified the topography and mapped the change in elevation (cut and fill), slope, aspect, curvature, and landform.

We CNC routed the base and mold for the model from MDF based on a DEM derived from a 2013 lidar survey. We used the base and the mold to cast a sculptable sand model. Each side of the 1:1500 scale model is 50 cm and represents 750 m. The model is two times vertically exaggerated and has 44 mm of relief which represent 33 m of elevation change.

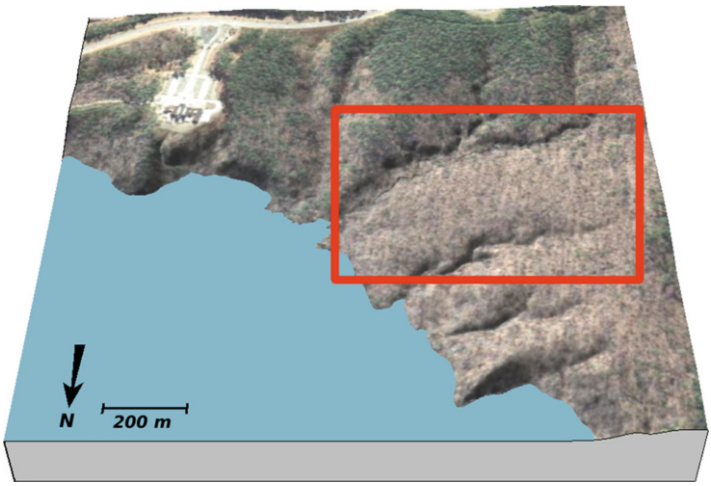

Fig. 4.6 Lake Raleigh Woods with our study site highlighted

We sculpted the sand model to grade a road and building sites. With Tangible Landscape we were able to analyze these modifications and understand how we had changed the slope, aspect, shape, and volume of the landscape.

4.2.2 Basic Workflow with DEM Differencing

To explore the impact of the new development we graded building sites and a road for the future neighborhood. We start with the original molded sand model, scan it and save the scan under different name so that it is not overwritten by the next scan. Then we compute and display 1 m countours:

```
g.rename raster=scan,scan_before
g.region raster=scan_before
r.contours input=scan_before output=contours_1m step=1
```

Figure 4.7a shows the terrain conditions before grading. While sculpting the model, we can pause the continuous scanning or keep it running to get ongoing feedback on our design as we progress.

We can visualize the magnitude and extent of our changes by differencing the scanned elevation before and after the change and then assigning a diverging color scheme to helps us distinguish between cut and fill (Fig. 4.7d):

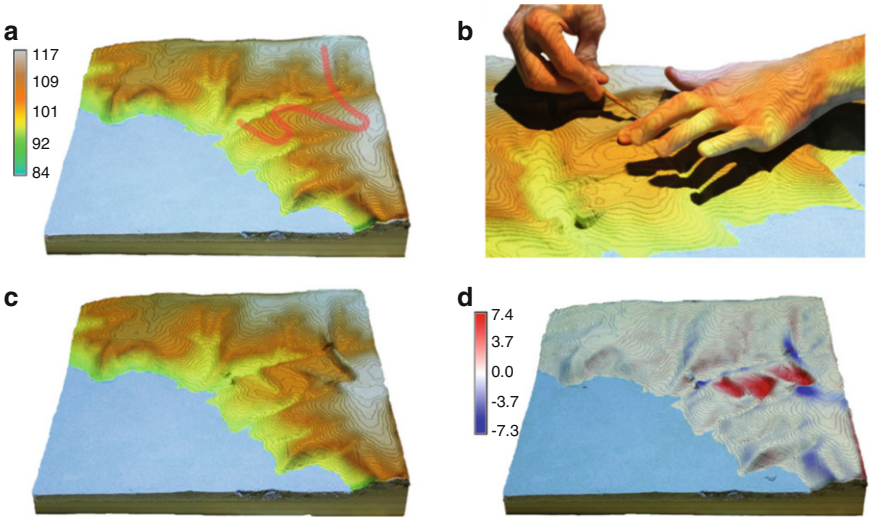

Fig. 4.7 Basic workflow: (**a**) the scanned elevation with contours, a *red curve* indicates the planned road; (**b**) modifying the terrain; (**c**) the finished design; (**d**) and the difference in meters between the elevation before and after grading, *red* representing soil removal (cut) and *blue* representing soil deposition (fill)

```
g.rename raster=scan,scan_after
r.mapcalc "diff = scan_after - scan_before"
r.colors -n map=diff color=differences
```

Because the scans are noisy, they do not align precisely in vertical direction. When computing difference between scans, any vertical shift is undesirable and should be minimized. The following workflow uses linear regression to match one scan to the other on the assumption that there has only been a moderate change in elevation. The linear regression of the form $y = a + bx$ estimates the vertical shift a and scale coefficient b (which should be close to 1) for adjusting the scan. Here we define a function in Python for adjusting the scan to match the other; we will refer to this function throughout the book:

```
def adjust_scan(scan_before, scan_after, scan_adjusted):
   coeff = gscript.parse_command('r.regression.line',
       mapx=scan_after, mapy=scan_before, flags='g')
   gscript.mapcalc("{scan_adjusted} = {a} + {b} *
       {scan_after}".format(scan_adjusted=scan_adjusted,
       a=coeff['a'], b=coeff['b'], scan_after=scan_after))
```

Here we use the function for adjusting scans to compute the difference:

```
adjust_scan('scan_before', 'scan_after', 'scan_adjusted')
gscript.mapcalc("diff = scan_adjusted - scan_before")
gscript.run_command('r.colors', flags='n', map='diff',
    color='differences')
```

4.2.3 The Impact of Model Changes on Topographic Parameters

In our case study the new neighborhood is planned on an east-facing hillside with a gentle slope of about 5° (Fig. 4.8). The slope values are steep on two sides of the hillside where it changes into parallel valleys; the profile curvature increases at the edge of the valleys (Fig. 4.9). The profile curvature on the hillside varies between positive and negative values denoting convex (ridges) and concave shapes (valleys).

We computed the slope and aspect using the *r.slope.aspect* module based on the standard polynomial approximation using 3×3 window, see Eqs. (4.9)–(4.14). When computing curvature, we use a more flexible method with a larger window (21×21 grid cells) to capture the larger scale morphology associated with the main valleys and ridges at our study site. This method is implemented in the module *r.param.scale* and we used this module to compute profile and tangential curvature. To visualize the variability in the curvature values we used a dedicated color ramp with divergent scheme:

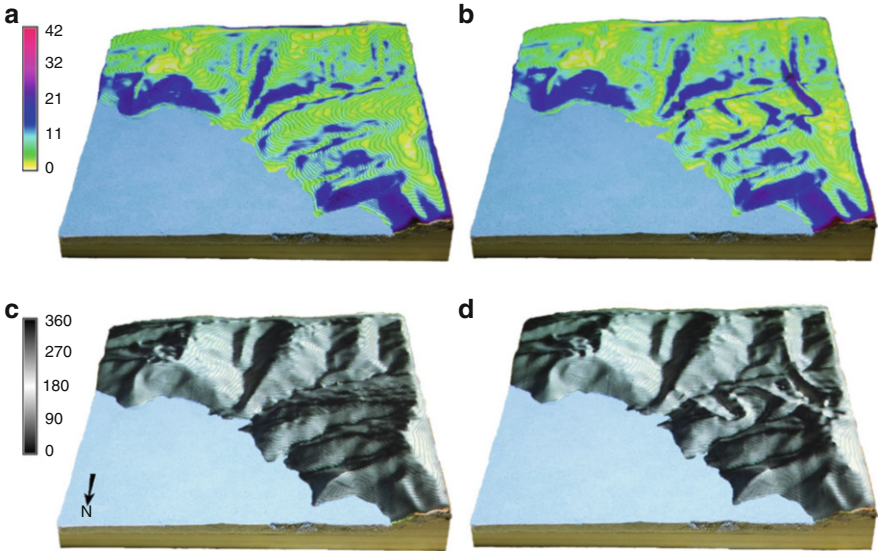

Fig. 4.8 Slope (**a**, **b**) and aspect (**c**, **d**) in degrees computed from the scanned model before (**a**, **c**) and after (**b**, **d**) modification

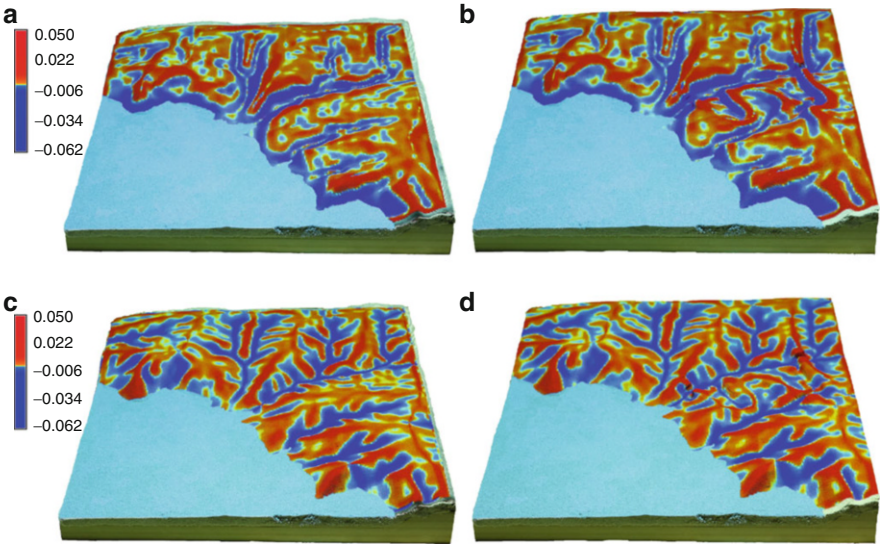

Fig. 4.9 Profile (**a**, **b**) and tangential (**c**, **d**) curvature computed from the scanned model before (**a**, **c**) and after (**b**, **d**) modification

```
r.slope.aspect elevation=scan aspect=scan_aspect \
    slope=scan_slope
r.param.scale input=scan output=scan_pcurv size=21 method=longc
r.param.scale input=scan output=scan_tcurv size=21 method=crosc
r.colors map=scan_pcurv color=curvature
r.colors map=scan_tcurv color=curvature
```

To build a new road with gentle slopes we constructed a series of switchbacks with low areas raised on embankments. As a result there are steep slopes beside the road on the faces of the embankments (Fig. 4.8). We also flattened parts of the hillside as building sites. When we analyzed the curvature of the modified landscape the flat roads were mapped as concave areas with high curvature due to the larger neighborhood size used in the analysis (Fig. 4.9).

4.2.4 Changing Landforms

The new development would change the landforms creating new ridges and valleys. We explored these changes by running the *r.geomorphon* add-on module which can be installed using *g.extension*. We can tune the results with parameters such as *search* and *skip* that determine the appropriate scale of analysis. By increasing these parameters we can skip small terrain variations in order to capture larger landforms:

```
g.extension r.geomorphon
r.geomorphon dem=elevation forms=landforms search=16 skip=6
```

In the resulting landform classification the new road has formed a major ridge (Fig. 4.10).

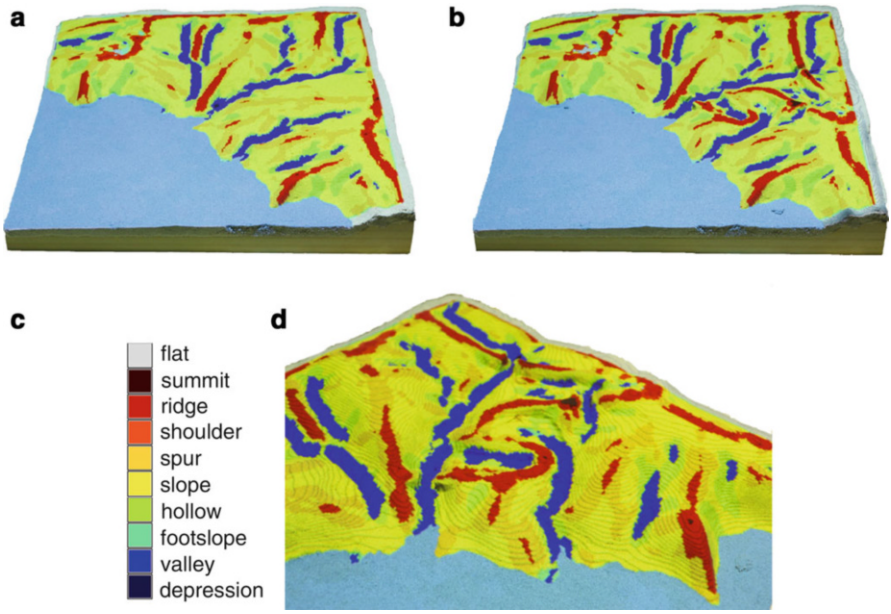

Fig. 4.10 Landform classification: (**a**) before development, (**b**) after development with the new elevated road classified as a ridge, (**c**) legend and (**d**) a detailed view of the modification done in Fig. 4.7

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